

DATA PROFILING LOG — Credit Risk

Project Date: 3rd April 2026

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Data:

Total Rows: 32,852

Total Columns: 12

Columns with nulls:

- Nulls in loan_int_rate: 3,115 rows
- Nulls in person_emp_length: 894 rows

Action: Remove all null rows in Power Query

loan_status values: 0 – No default and 1- Default

What we are building: A 3-tool portfolio project simulating a bank's retail loan portfolio risk monitoring system.

The story we're telling: "A retail bank has 5,000 loan accounts. A job as the analyst is to clean the data, score each customer's risk level, identify the high-risk segment, and present a dashboard that helps the risk committee make decisions."

This is a real Monday morning task at companies like JP Morgan, Citi, HSBC.

Observations:

Observation 1:

- High Risk and Medium Risk tiers dominate the portfolio.
- Low Risk accounts appear to be the minority.

Hypothesis: This dataset represents a subprime lending portfolio — customers who were approved despite elevated risk profiles. This will likely show a higher than average default rate.

Observation 2:

- 22% default rate against industry average of 2-5%.
- Portfolio is significantly distressed.
- Average interest rate of 11.04% confirms subprime positioning.
- Immediate risk review recommended.

Observation 3:

- Medium Risk dominates at 45% — this is borderline acceptable.
- However High Risk + Critical Risk combined = 28.81% of portfolio.
- Nearly 1 in 3 customers is High or Critical Risk.

- Combined with 22% default rate this suggests poor credit underwriting standards at loan origination.

Observation 4:

- Top 3 loan purposes are Education, Medical and Venture.
 - These are non-collateralised, necessity-driven loans.
 - Customers borrowing for education and medical expenses are likely under financial stress already — higher default probability than home improvement or debt consolidation.
 - Venture loans indicate risk-seeking borrower behaviour. Combined with 22% default rate this portfolio requires immediate segmentation by loan purpose vs default rate.
- Total Customers: 28,632
 - Portfolio Value: £276 Million
 - Avg Interest Rate: 11.04%
 - Default Rate: 22%
- Risk Distribution:
 - Medium Risk: 45.25%
 - High Risk: 27.35%
 - Low Risk: 25.94%
 - Critical Risk: 1.46%
- Defaults: 6,000 customers
 - Non-Defaults: 22,000 customers
- Top Loan Purposes:
 - 1. Education- 5,700
 - 2. Medical - 5,300
 - 3. Venture- 5,000
 - 4. Personal/Debt Consolidation
 - 5. Home Improvement

Observation 5:

- 55% of customers fall in medium income band (30k-70k).
- Only 8.7% are Very High-income earners.
- Combined with 22% default rate this suggests the portfolio is heavily exposed to middle income borrowers who are financially stretched — most vulnerable to interest rate rises and economic downturns.

Data correction: Income Band formula corrected during build — Medium threshold fixed from 7,000 to 70,000.

Observation 6 — KEY FINDING:

- Risk Tier classification validates as a strong default predictor.
- Critical Risk customers default at 69% — nearly 7 in 10.
- High Risk customers default at 40% — nearly 1 in 2.
- Low Risk customers default at only 9% — well controlled.
- Tightening credit standards for interest rates above 13% could reduce portfolio default rate significantly.

Recommendation: Bank should immediately review lending criteria for High and Critical Risk segments.

Observation 7:

- Debt Consolidation: 28% default rate
- Medical: 27% default rate
- Home Improvement: 26% default rate

Counterintuitive finding — Debt Consolidation has highest default rate at 28%, not Education or Venture as initially hypothesised.

Interpretation: Customers consolidating debt are already financially distressed at loan origination. They are borrowing to pay off existing debt — a sign of cascading financial stress.

These customers have highest default risk despite appearing to be responsible borrowers.

Recommendation: Debt Consolidation applications should trigger enhanced credit scrutiny and lower loan-to-income ratios at approval stage.

Observation 8:

- Employment stability shows expected inverse relationship with default rate for 0-15 years.
- Unexpected spike at 25+ years suggests mid-to-late career financial stress — possibly retirement planning pressure, dependent family expenses, or accumulated debt burden.

Statistical note: Data points beyond 35 years employment have small sample sizes — findings not statistically reliable at extreme ends.

Recommendation: Employment years should be used as a secondary risk indicator, not primary. Most predictive in the 0-5 year range.

Observation 9:

Perfect progressive relationship between Risk Tier and average loan amount. Critical Risk customers borrow 70% more than Low Risk (£14,600 vs £8,600).

Combined with 69% default rate for Critical Risk:

These customers are borrowing the most AND repaying the least.

This represents the highest loss exposure in the portfolio.

Recommendation: Implement loan amount caps for High and Critical Risk customers. Maximum loan amount should be calibrated against income and credit history, not just credit score.

Observation 10:

Credit history length shows negligible predictive power for default risk in this dataset.

Flat trend from 0-20 years suggests length alone is insufficient as a risk indicator.

Limitation: Dataset lacks credit score quality metrics which would normally accompany history length in a real banking system. Slight uptick after 20 years consistent with Observation 8 — age-related financial stress pattern emerging across multiple variables.

Recommendation: Credit history length should be deprioritised as a standalone risk indicator. Replace with credit score or payment delinquency count for stronger predictive power.

Observation 11 —

CRITICAL FINDING:

- Loan Grade shows perfect progressive default relationship from Grade A (lowest) to Grade G (highest).
- This cross-validates our Interest Rate based Risk Tier classification from Page 2.
- Two independent risk models agree on the same conclusion — increases confidence in our risk assessment framework.
- In a real banking environment this alignment would justify using Interest Rate as a viable proxy for internal loan grade when grade data is unavailable.

Observation 12 —

MOST IMPORTANT BUSINESS FINDING:

- Counterintuitive risk concentration discovered. Critical Risk has highest default RATE at 69% but only £6M total exposure —manageable loss scenario.
- Medium Risk has only 16% default rate BUT £126M exposure — largest systemic risk in the portfolio.
- A 10% deterioration in Medium Risk repayment behaviour would cost the bank £12.6M — 3x the entire Critical Risk portfolio value.

Recommendation: Immediate portfolio health monitoring program for Medium Risk segment. Early intervention and payment support schemes should be prioritised for this segment over Critical Risk due to scale of financial exposure.

Observation 13 — ACTIONABLE POLICY FINDING:

Clear default rate inflection point identified at 12% interest rate threshold.

Below 12%: default rate 9-15% — within manageable range

Above 12%: default rate jumps to 31-86% — unsustainable

Recommended Credit Policy Change:

Loans above 12% interest rate should require:

1. Enhanced credit committee approval
2. Lower loan-to-income ratio limits
3. Mandatory collateral assessment
4. Quarterly portfolio review trigger

This single policy change could reduce portfolio default rate from 22% to approximately 12% by restricting the highest risk approvals.

Observation 14:

- Portfolio heavily concentrated in young borrowers peaking at age 23.
- Young borrowers + Education/Medical loan purposes +
- Low financial buffers = explains elevated default rate.

Recommendation: Age-adjusted credit scoring should be implemented. Young borrowers need income trajectory assessment not just current income snapshot.

Observation 15:

- 92% of borrowers either rent or have a mortgage — meaning virtually no customers have unencumbered assets.
- This portfolio has extremely low collateral coverage.
- In a default scenario the bank has almost no assets to recover losses against.

Recommendation: Collateral assessment should be mandatory for all High and Critical Risk applications.

Secured lending products should be developed for mortgage and rent paying customers to reduce unsecured exposure.

Observation 16 —

COMPLETE BORROWER PROFILE:

- Dominant customer segment is young urban renters aged 20-25 with under 3 years employment history borrowing for education and medical necessities.
- This segment combines every major default risk factor: youth, employment instability, no collateral, necessity-driven borrowing.
- Portfolio was likely built targeting underserved young professionals — high social value but requires specialist risk management approach.